
MISSION-CENTRIC REQUIREMENTS ANALYSIS OF MODEL PREDICTIVE CONTROL IN SPACECRAFT RENDEZVOUS AND PROXIMITY OPERATIONS

TECHNICAL REPORT — ADRIOS / CLEARSPACE-1 APPLICATION

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ABSTRACT

Rendezvous and proximity operations (RPO) missions are commonly designed as a sequence of distinct phases each governed by its own sample time, constraint set, and cost structure, with the translation from mission-level requirements to model predictive control (MPC) design parameters typically performed ad hoc. This report takes a first step toward a systematic, traceable mapping between mission requirements and MPC design choices, using the ADRIOS / ClearSpace-1 active debris removal mission as a case study across three sequential operations. Its central hypothesis is that Economic MPC (EMPC), in which the stage cost reflects propellant consumption directly, is structurally better suited to proximity operations than fixed-point regulation or reference tracking, because it can exploit the natural, fuel-free relative-orbital-motion trajectories admitted by the Clohessy–Wiltshire–Hill (CWH) dynamics instead of fighting them. We show that a pure-fuel stage cost is fundamentally degenerate for this class of problem as the closed-loop optimum is not strictly dissipative with respect to any single target state or orbit, and analyse, operation by operation, which terminal ingredient (a terminal region, a terminal cost, a terminal equality, or a stage-cost regularisation) is required to repair this degeneracy, at what fuel and time cost, and under what formal guarantee. Across all three operations the economic controllers cut fuel consumption by roughly 40–99% relative to standard quadratic-tracking MPC, at the cost of longer manoeuvre times, and we synthesise the results into a cross-scenario table of mission requirements, MPC ingredients, and their residual limitations.

Keywords Economic Model Predictive Control · Spacecraft Rendezvous · Proximity Operations · Active Debris Removal · Dissipativity

List of Acronyms

ADR	Active Debris Removal
CAM	Collision Avoidance Manoeuvre
CLF	Control Lyapunov Function
ConOps	Concept of Operations
CW	Clohessy–Wiltshire
CWH	Clohessy–Wiltshire–Hill
DARE	Discrete Algebraic Riccati Equation
ECI	Earth-Centred Inertial
EMPC	Economic Model Predictive Control
ESA	European Space Agency
FAP	Final Approach Point
FKP	Formation Keeping Point
GNC	Guidance, Navigation and Control
IPP	Initial Proximity Point
KOS	Keep-Out Sphere
LEO	Low Earth Orbit
LEOP	Launch and Early Orbit Phase
LQR	Linear Quadratic Regulator
LTI	Linear Time-Invariant
MPC	Model Predictive Control
NMC	Natural Motion Circumnavigation
QP	Quadratic Program(ming)
RPO	Rendezvous and Proximity Operations
ZOH	Zero-Order Hold
ISS	International Space Station
SRP	Solar Radiation Pressure

Nomenclature

Symbol	Description	Unit
n	Target mean motion	rad/s
T	Orbital period	s
Δt	Sample time	s
N	Prediction horizon length	steps
x, y, z	Hill-frame relative position (radial, along-track, cross-track)	m
$\dot{x}, \dot{y}, \dot{z}$	Hill-frame relative velocity components	m/s
$\mathbf{x}$	State vector $[x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top$	—
u	Control input (specific thrust)	m/s ²
$u_{\max}$	Maximum thrust / acceleration bound	m/s ²
A_c, B_c	Continuous-time system matrices	—
A_d, B_d	Discrete-time (ZOH) system matrices	—
Q, R	Stage-cost weighting matrices (state, input)	—
$Q_{\text{dare}}, R_{\text{dare}}$	DARE design weights	—
P	Terminal cost matrix (DARE solution)	—
K	Stabilising feedback gain	—
ℓ	Stage cost function	—
λ	Storage function (dissipativity)	—
$\rho(\cdot)$	Class- $\mathcal{K}_\infty$ function (dissipativity margin)	—
ρ_c	Stage-cost regularisation weight	—
ε	Regularisation weight, augmented stage cost	—
x_s, u_s	Optimal economic steady state	—
x_{ref}	Generic reference/target state	—
x_{dock}	Docking-port target state (Op. 3)	—
x_{switch}	Handover-point target state (Op. 1)	—
e	Error state, $e = x - x_{\text{ref}}$	—
Δv	Total characteristic velocity (fuel usage)	m/s
r_{KOS}	Keep-out sphere radius	m
ϕ_0	Initial phase angle of NMC orbit	rad
y_c	Along-track drift centre of NMC orbit	m
b	NMC ellipse radial semi-axis (used interchangeably with ρ in Op. 2/3 discussion); along-track semi-axis is $2b$	m

1 Introduction

Active debris removal (ADR) missions are structured as a sequence of distinct proximity operations. A far-range closing approach, an inspection or fly-around phase, and a final docking or capture manoeuvre [1, 2]. Model Predictive Control (MPC) is well suited to this kind of mission because it handles input and state constraints explicitly over a receding horizon [3, 4], and because the same underlying optimisation problem can, in principle, be re-parameterised from one operation to the next rather than re-designed from scratch. MPC has accordingly become a leading approach for constrained spacecraft rendezvous and proximity operations [5, 6, 7, 8].

In practice, however, the translation from mission-level requirements like orbital regime, approach-corridor geometry, fuel budget and keep-out zones to MPC design parameters like horizon length, cost structure, terminal ingredients and constraint formulation is typically performed ad hoc, on a per-mission and per-phase basis, without a systematic methodology connecting the two.

This is the gap the thesis tries to close. The eventual goal is a requirements framework that maps mission parameters onto MPC design choices in a traceable, reusable way, developed across an entire mission sequence rather than a single isolated manoeuvre.

The present report is a first step towards it. Its central working hypothesis is that Economic MPC (EMPC) [9, 10, 11, 12], in which the stage cost reflects the physical resource of interest directly, e.g. propellant, is structurally better suited to proximity operations than fixed-point regulation or reference-tracking MPC, because it can exploit natural relative-orbital-motion dynamics instead of fighting them.

The ADRIOS / ClearSpace-1 mission [13] is used here to derive the framework. The mission serves two purposes:

1. Expose which problems arise when an MPC is constructed for a specific proximity operation
2. Identify where EMPC is useful, why it is useful and what mission requirements creates that need

This report documents the current implementation status of this first application to the ADRIOS / ClearSpace-1 Concept of Operations (ConOps) [13], treated as three sequential operations.

1. Operation 1 : far-range closing
2. Operation 2 : fly-around by using the Natural Motion Circumnavigation arising from the CW-equations [14]
3. Operation 3 : final docking

Section 2 lays out the above sequence and the requirements each phase imposes; Section 3 establishes the rendezvous dynamics and how they have nice properties for one specific Rendezvous and Proximity Operation (RPO) mission; Section 4 argues why MPC and especially EMPC is the right tool for these dynamics; Section 5 introduces the shared EMPC formulation and design axis used across all three operations; Sections 6.1–6.3 report the implementation and results for each operation in turn; Section 7 draws the results together into a cross-scenario requirements comparison; and Section 8 summarises what is established and what remains open.

2 Mission Overview & Scenarios

2.1 ADRIOS ClearSpace-1

ADRIOS ClearSpace-1 is the first European mission to demonstrate in-orbit rendezvous, capture, and removal of an existing non-cooperative space object from low Earth orbit [13]. The selected client is PROBA-1, an ESA microsatellite (launched 2001, approx. 94 kg, $60 \times 60 \times 80$ cm) in a Sun-synchronous orbit at roughly 400 km altitude. The mission is developed by OHB System AG and ClearSpace SA under ESA's Space Safety Programme, with launch expected in 2028.

2.2 Concept of Operations

After launch, parallel commissioning and phasing bring the servicer to a switch point approximately 30 km behind PROBA-1. At this point absolute navigation hands over to relative GNC, initiating the rendezvous and proximity operations (RPO) phase. This far-to-close-range phasing follows the classical rendezvous sequence [1, 15]. The full ADRIOS ConOps comprises the following sequential phases [13]:

1. **Closing.** Far-range transfer from 30 km to a few hundred metres, using passively safe impulsive manoeuvres and angles-only optical navigation. A ranging device supplements the camera in the final portion.

2. **Fly-around.** The servicer enters NMC orbits around the client at 50–100 m, observing and characterising PROBA-1 with mid-range navigation. The same guidance concept as closing is used; range measurements augment the camera for improved robustness.
3. **Proximity manoeuvring.** The servicer transfers to the Formation Keeping Point (FKP) at 150 m along-track, hops to the Initial Proximity Point (IPP), and then advances to the Final Approach Point (FAP) using close-range navigation and motion synchronisation to track PROBA-1’s rotation. A rehearsal of the full sequence precedes each capture attempt; the mission is sized for three capture attempts with one Collision Avoidance Manoeuvre (CAM) budgeted per attempt.
4. **Capture.** Once PROBA-1 is inside the capture volume, GNC hands off to the capture system, which closes four robotic arms around the client. After securing the client, the stack detumbles and a perigee-lowering manoeuvre initiates deorbit within five years.

2.3 Scope of This Report

This report addresses the RPO GNC problem for a simplified version of the first three phases of the above sequence. The proximity manoeuvring phase (FKP $\rightarrow$ FAP and motion synchronisation) requires six-degrees-of-freedom pose-tracking control and is outside the scope of the present work. Thus, Operation 3 deals with a simplified terminal approach, which takes the chaser from fly-around to the targets eventual docking port, where the capture system takes over. The three operations studied here are treated not as independent test cases but as one continuous mission sequence in which the same design translation (mission requirement to MPC ingredients) is instantiated three times under changing range and dynamics regime. Table 3 lists the sequence. Each row corresponds to a single design instance, in which the sample time, horizon and terminal ingredient are selected according to that phase’s requirements. Δt is here the ZOH discretisation step of the prediction model and the rate at which a new control input is computed and applied.

In Operation 1, $\Delta t = 120$ s is loose relative to the CW timescale ($T = 5545$ s) because far-range closing involves slowly evolving relative motion and a wide (50 m) convergence tolerance, so finer sampling would add computational cost without improving control authority. In Operation 2, $\Delta t = T/92 \approx 60.27$ s is fixed by the requirement that the horizon land exactly on one full orbital period, $N\Delta t = T$, since the periodic terminal equality (19) needs the predicted trajectory to close the NMC orbit exactly rather than approximately. In Operation 3, $\Delta t = 10$ s is an order of magnitude finer because the docking-approach line-of-sight cone and the sub-metre terminal tolerance both tighten as range decreases, and a coarser sample would let the chaser drift outside the corridor between updates. The horizon length N is likewise set by a different requirement in each case (see Section 6)

Table 3: Operations covered in this report (from the ADRIOS ConOps), and the mission requirement that drives each MPC design choice.

Op	Scenario / Mission requirement	Range	$\Delta t / N$	Horizon ($N\Delta t$)	MPC design choice
1	Far-range closing / reach box around 300 m switch point	10 km $\rightarrow$ 300 m	120 s / 20	2400 s (≈ 40 min)	Pure fuel + ℓ_∞ terminal ball; + $\rho_c \ e\ ^2$ regularisation; + CLF terminal cost ($M_1 \succeq 0$)
2	NMC fly-around / acquire NMC orbit	300 m $\rightarrow$ NMC orbit	≈ 60.27 s / 92	5545 s ($= T \approx 1.54$ h)	Pure fuel + periodic terminal eq. $x_N = x_{\sigma(k)}^*$; + $\varepsilon \ x - x_k^*\ ^2$ phase-sync
3	Final docking / reach docking-port vicinity	150 m $\rightarrow$ <1 m	10 s / 24	240 s ($= 4$ min)	Pure fuel + CLF terminal cost; + hard terminal eq. $x_N = x_{\text{dock}}$

3 Rendezvous Dynamics

3.1 Rendezvous Mechanics and the CW Equations

Spacecraft rendezvous requires driving the relative state of a chaser vehicle to a desired position/velocity relative to a target. The full two-body problem in an Earth-Centred Inertial (ECI) frame is nonlinear. For small relative separations and near-circular reference orbits, the relative dynamics linearise to the Clohessy–Wiltshire (CW) equations [16, 17, 14]:

$$\ddot{x} - 2n\dot{y} - 3n^2x = u_x, \quad (1)$$

$$\ddot{y} + 2n\dot{x} = u_y, \quad (2)$$

$$\ddot{z} + n^2z = u_z, \quad (3)$$

with state vector $[x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top$. These form a 6-state, 3-input linear time-invariant (LTI) system.

The CW equations rest on a circular target orbit (constant n), unperturbed two-body gravity (no J_2 , drag, or SRP), and a first-order linearisation of the differential gravity field valid only for relative separations small compared to the orbital radius [1, 14]. It further assumes the chaser’s mass and thrust are too small to perturb the target’s trajectory, and that the Hill frame rotates at the constant rate n . All three operations in this report satisfy these conditions comfortably so the CW model is expected to hold with negligible mismatch error throughout.

A key structural property is visible directly in the equations as x and y are coupled through the Coriolis terms ($2n\dot{y}$, $2n\dot{x}$), while z decouples entirely and behaves as a simple harmonic oscillator at frequency n . The in-plane (x - y) motion combines a bounded oscillatory mode with a secular along-track drift term that grows linearly in time. The CW system is therefore not asymptotically stable. Only under a specific initial-condition constraint, the drift-free condition introduced in Section 3.2, does the secular term vanish, leaving a closed, periodic orbit (Natural Motion Circumnavigation, NMC). More generally, the unforced ($u \equiv 0$) response is dominated by the along-track drift [14, 1]. This structural instability is the fundamental challenge for rendezvous control but it is also, for EMPC specifically, a resource to be exploited.

3.2 Natural Motion Circumnavigation

Among the unforced solutions of the CW equations (Section 3.1), one family is of particular operational value. The closed, periodic relative orbits known as Natural Motion Circumnavigations (NMC). Because they require no propellant to maintain, they are the natural target regime for the fly-around inspection phase [14, 1]. The along-track solution of the CW equations carries a secular term ($6nx_0 + 3\dot{y}_0$) that grows linearly in time and drives the chaser away from the target. Bounded motion therefore requires this term to vanish, which gives the drift-free condition

$$\dot{y}_0 = -2nx_0. \quad (4)$$

When (4) holds, the chaser executes a closed, periodic orbit relative to the target. When it is violated (e.g. $\dot{y}_0 = 0$), a secular along-track drift accumulates, since the y -dynamics carry no restoring force.

With the drift-free condition and $\dot{x}_0 = 0$, the in-plane motion reduces to

$$x(t) = x_0 \cos nt, \quad (5)$$

$$y(t) = -2x_0 \sin nt + y_0, \quad (6)$$

a closed ellipse with radial semi-axis $|x_0|$ and along-track semi-axis $2|x_0|$. The geometry acquired in Operation 2 (Section 6.2).

4 Motivation: Why (Economic) Model Predictive Control?

Classical regulation techniques, like unconstrained PD control or LQR do not represent mission constraints explicitly. Rendezvous and Proximity Operations (RPO) is therefore a poor match for these techniques, because nearly every requirement a proximity operation has to satisfy is some kind of constraint, e.g. keep-out zones and passive-safety envelope constrain the state, thrust and total- Δv budgets bound the input and line-of-sight cones bound the geometry between chaser and target [2, 5]. This is where MPC yields advantages over classical regulation techniques. It poses the control problem as a constrained, finite-horizon optimisation, re-solved on a receding horizon. This allows the

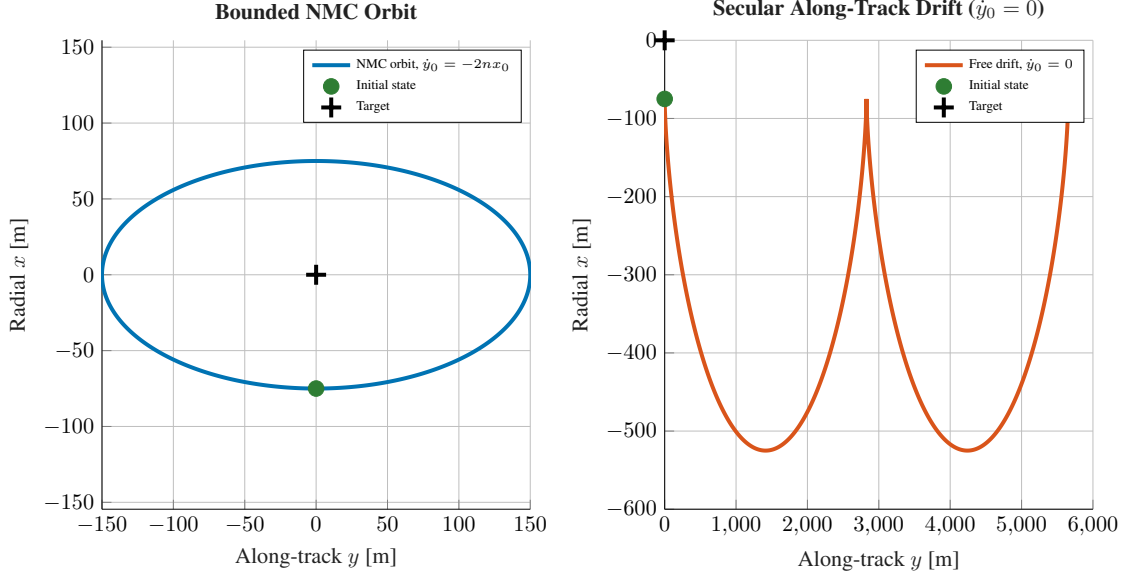


Figure 1: Bounded 2:1 NMC ellipse in the Hill frame under the drift-free condition $\dot{y}_0 = -2n x_0$. The chaser traces a closed, periodic orbit about the target with no along-track drift. When this condition is violated (e.g. $\dot{y}_0 = 0$), the uncanceled secular term causes the along-track coordinate y to grow linearly in time, and the trajectory drifts away from the target instead of closing.

inclusion of state and input constraints in the problem formulation [3, 4]. Therefore MPC is the natural candidate for proximity operations as you can also see in the extensive literature published in the last years [5, 6, 7, 8].

A second reason for using MPC in RPO specifically is the sequential structure of the RPO missions. Throughout their phases, far-range closing to docking, the sample time, active constraint set, and stage/terminal cost weighting all change from phase to phase, driven by the tightening safety margins and update-rate requirements as range decreases. What remains fixed across this entire sequence is the underlying relative-motion model (the CWH dynamics themselves), the condensed QP structure, and the solver architecture used to solve it [1, 2]. Weiss et al. demonstrated exactly this separation. A rendezvous-phase QP with a long sampling period is switched to a docking-phase QP with a shorter sampling period and a different constraint set, while the prediction model and QP solver architecture are carried across the switch unchanged [6].

This report exploits exactly this re-parameterisation capability where the same underlying optimisation problem is carried across operations, with only the horizon, constraint set, and cost structure adjusted as the mission moves from one phase to the next.

4.1 Regulation, Reference Tracking, and Economic Cost

Given that MPC handles constraints natively, the remaining design freedom is the cost function, and here the RPO-MPC literature reduces to two classical choices. The first, and by far the most common [5, 6, 2], penalises the deviation of state and input from a fixed equilibrium setpoint,

$$\ell_{\text{reg}}(x, u) = \|x - x_s\|_Q^2 + \|u - u_s\|_R^2,$$

i.e. pure fixed-point regulation. The second replaces the constant setpoint with a precomputed, time-varying reference x_k^* and follows it in a phase-synchronised manner,

$$\ell_{\text{track}}(x_k, u_k) = \|x_k - x_k^*\|_Q^2 + \|u_k - u_k^*\|_R^2.$$

Both retain a quadratic state penalty and differ only in whether the reference is a point or a trajectory. In this report the closing and docking phases are point-stabilisation problems, for which ℓ_{reg} is the natural baseline. The fly-around instead targets a phase-synchronised point on the NMC. Exactly the trajectory the economic cost is meant to exploit.

An economic cost is especially interesting for RPO because the relative orbital dynamics admit trajectories that consume no fuel at all yet do not sit at any fixed point. The CW state-transition matrix has purely imaginary eigenvalues, and the resulting sinusoidal closed-form solutions yield bounded, periodic, thrust-free trajectories (NMCs) along which a chaser can remain in the target's vicinity indefinitely without expending propellant [2]. This exposes the

weakness of both quadratic designs. A fixed-point regulator asked to hold a single point on (or near) such an orbit is, by construction, fighting the natural drift and burning fuel to cancel free motion, and while a phase-synchronised tracker can capture the NMC, it does so only if the fuel-free reference x_k^* is known and supplied in advance.

This is the direct motivation to go with an economic cost.

4.2 Economic MPC

Economic MPC (EMPC) lets the cost encode the physical resource of interest directly [9, 12]. For a thrusting spacecraft the propellant expended over a manoeuvre is the accumulated velocity increment,

$$\Delta v = \int \|u(t)\| dt,$$

so an exact minimum-propellant objective is the 1-norm stage cost $\ell \propto \|u\|$. In this work we instead adopt the quadratic control-effort surrogate

$$\ell_{\text{eco}}(u) = \|u\|_R^2,$$

a smooth, strictly convex proxy that keeps each receding-horizon step a well-conditioned QP and yields continuous thrust profiles, at the price of not minimising Δv exactly. What matters for the present argument is the feature both share, namely the absence of any state-tracking term. Reported Δv values are consequently evaluated a posteriori as $\int \|u\| dt$, not as the optimised quantity. By omitting the state penalty the optimiser is free to exploit whichever admissible motion minimises cumulative effort. For a system whose unforced dynamics already contain the fuel-free NMC trajectories, this is precisely the freedom a quadratic tracking or regulation cost forfeits.

This freedom, however, forfeits the standard stability certificate. The economic stage cost carries no state penalty, so it does not vanish only at the desired steady state and grow away from it, and it therefore cannot serve as a Lyapunov function. The Riccati-based terminal ingredient that certifies quadratic MPC through its equivalence to the infinite-horizon LQR [5, 6] no longer applies. Indeed, an economic stage cost need not be positive definite with respect to the target at all [12]. Recovering a convergence guarantee thus calls for a different structural property. A well-established route is the concept of strict dissipativity [18].

A system $x^+ = f(x, u)$ with stage cost ℓ is strictly dissipative with respect to an optimal steady state (x_s, u_s) if there exists a storage function $\lambda : \mathbb{X} \rightarrow \mathbb{R}$ and a class- $\mathcal{K}_\infty$ function ρ such that the rotated stage cost

$$\tilde{\ell}(x, u) = \ell(x, u) - \ell(x_s, u_s) + \lambda(x) - \lambda(f(x, u)) \geq \rho(\|x - x_s\|) > 0 \quad (x \neq x_s),$$

for all admissible (x, u) [18, 9, 10]. Because $\tilde{\ell}$ differs from ℓ only by a telescoping term and a constant, both yield the same optimal inputs, yet $\tilde{\ell}$ is positive definite in x about x_s , so the optimal value function of the rotated problem is a valid Lyapunov function [10]. This restores the standard MPC convergence argument for the economic problem, which is why constructing the storage function λ is the central analytical step of any EMPC stability claim.

For the CW system under a pure propellant cost $\ell(x, u) = \|u\|$ (or $\|u\|^2$), this necessity is not given due to structural reasons. The economic cost is minimised, with value zero, by $u = 0$. Since the entire V-bar axis $\{x = [0, y, 0, 0, 0, 0]\}$ consists of relative equilibria, and because every NMC is a bounded, thrust-free periodic solution of zero stage cost, the optimiser is thus presented with a continuum of zero-cost steady states and periodic orbits. With all of them sharing the minimal average cost, namely $u = 0$. Strict dissipativity, however, demands a strictly positive margin $\rho(\|x - x_s\|)$ that grows as the state moves away from a single distinguished optimal steady state x_s . A continuum of equally optimal, zero-cost solutions yields that no storage function can render the rotated cost uniquely minimal at a single point.

The results of this degeneracy are obvious. A bare-fuel objective contains no mechanism for selecting the desired state or orbit. Thus the controller will drift towards whichever equilibrium is cheapest to reach (see Section 6.2).

Accordingly, the remainder of this report is about how to repair this degeneracy. The candidate routes all reintroduce, in one form or another, the “preference” that the bare economic cost lacks, e.g. hard terminal constraints that pin the target state or orbit directly, terminal costs that shape the value function near the target without forcing it exactly or a stage-cost regularisation that adds a small state-dependent term to recover a strict (or practical) dissipativity inequality. Section 5 formalises the shared machinery behind them, and Sections 6.1–6.3 examine, operation by operation, which route is appropriate and why.

5 MPC Formulation & Concepts

5.1 System Model

With state $x = [x, y, z, \dot{x}, \dot{y}, \dot{z}]^\top$ and control input u (specific thrust, i.e. acceleration), the continuous-time CWH dynamics are [17, 14]

$$\dot{x} = A_c x + B_c u, \quad A_c = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 3n^2 & 0 & 0 & 0 & 2n & 0 \\ 0 & 0 & 0 & -2n & 0 & 0 \\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix} \quad B_c = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (7)$$

where n is the target's mean motion. For the simulation, an ISS-like reference orbit was used, with $n = 0.001133$ rad/s and orbital period $T = 5545$ s.

ZOH discretisation at scenario-appropriate sample time Δt yields the discrete-time system

$$x_{k+1} = A_d x_k + B_d u_k, \quad A_d = e^{A_c \Delta t}, \quad B_d = \int_0^{\Delta t} e^{A_c \tau} d\tau B_c. \quad (8)$$

The terminal-cost matrix P is the solution of the Discrete Algebraic Riccati Equation (DARE)

$$P = Q + A_d^\top P A_d - A_d^\top P B_d (R + B_d^\top P B_d)^{-1} B_d^\top P A_d. \quad (9)$$

5.2 Condensed QP Formulation

All controllers in this report are implemented in condensed (single-shooting) form, in which the predicted states are eliminated and the quadratic program (QP) is solved over the input sequence alone [4, 19]. Substituting the discrete dynamics $x_{i+1} = A_d x_i + B_d u_i$ repeatedly over a horizon N gives the lifted prediction

$$\mathbf{X} = S_x x(k) + S_u U, \quad \mathbf{X} = [x_1^\top, \dots, x_N^\top]^\top, \quad U = [u_0^\top, \dots, u_{N-1}^\top]^\top, \quad (10)$$

with prediction matrices

$$S_x = \begin{bmatrix} A_d \\ A_d^2 \\ \vdots \\ A_d^N \end{bmatrix}, \quad S_u = \begin{bmatrix} B_d & 0 & \cdots & 0 \\ A_d B_d & B_d & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ A_d^{N-1} B_d & \cdots & A_d B_d & B_d \end{bmatrix}. \quad (11)$$

For a quadratic stage and terminal cost with block-diagonal weights $\bar{Q} = \text{blkdiag}(Q, \dots, Q, P)$ and $\bar{R} = I_N \otimes R$, and an optional stacked reference $\mathbf{R}$, the finite-horizon problem reduces to the strictly convex QP

$$\min_U \frac{1}{2} U^\top H U + f^\top U \quad \text{s.t.} \quad \underbrace{-u_{\max} \mathbf{1} \leq U \leq u_{\max} \mathbf{1}}_{\text{input}}, \quad \underbrace{A_{\text{in}} U \leq b_{\text{in}}}_{\text{state / geometric}}, \quad \underbrace{A_{\text{eq}} U = b_{\text{eq}}}_{\text{terminal}}, \quad (12)$$

with

$$H = 2(S_u^\top \bar{Q} S_u + \bar{R}), \quad f = 2S_u^\top \bar{Q} (S_x x(k) - \mathbf{R}), \quad (13)$$

where $\mathbf{R} = 0$ for pure regulation.

Any state constraint the chaser must respect at every time step, written generally as $C x_i \leq d$, can be rewritten purely in terms of the inputs U instead of the states, using the same substitution (11) that built the prediction in the first place. Collecting all such constraints together gives the compact form $A_{\text{in}} U \leq b_{\text{in}}$ used in the QP.

Two such constraints are used to shape the approach geometry. (i) An along-track half-space, $y_i \geq 0$, simply keeps the chaser on the correct side of the docking port throughout the approach. (ii) A line-of-sight cone of half-angle α around the +V-bar axis, $\sqrt{x_i^2 + z_i^2} \leq y_i \tan \alpha$, keeps the chaser inside a cone so the docking sensors always have a clear view of the target. This cone is a smooth, curved shape, which the QP solver cannot handle directly, so it is approximated by M flat faces,

$$\cos \theta_j x_i + \sin \theta_j z_i \leq \tan \alpha \cos(\frac{\pi}{M}) y_i, \quad \theta_j = \frac{2\pi j}{M}, \quad j = 0, \dots, M-1, \quad (14)$$

where the extra $\cos(\pi/M)$ factor shrinks the faces slightly inward so that the resulting polygon sits entirely inside the true cone, never outside it [6]. This way the approximation is conservative. Any trajectory satisfying the flat-face version also satisfies the real cone constraint.

The condition placed on the very last predicted state is built the same way. Pinning the chaser exactly to the docking state at the end of the horizon, $x_N = x_{\text{dock}}$, becomes an equation in U alone, while a softer version, only requiring the chaser to land somewhere inside a small box around the target at the end of the horizon, is written as $\|C_p x_N\|_\infty \leq r$. A perfectly round target region would again turn the problem into a curved, more expensive one to solve, so a box is used instead, chosen so it always sits inside the true target region.

The QP is solved with MATLAB's quadprog [20]. Only the first block $u^*(k) = U_{1:n_u}^*$ is applied and the horizon recedes one step. Real-time and embedded implementations of this dense QP are well established [21, 22]. This construction is shared by every controller below, which differ only in the stage cost, terminal ingredient, reference, and active constraint set.

5.3 Economic MPC Formulation

Every economic controller sets $Q = 0$ in the condensed QP of Section 5.2, so the stage cost $\ell(x, u) = \|u\|^2$ reflects propellant directly. The structural consequences of this choice, the dissipativity gap and the design axis for closing it, are developed in the context of Operation 1 where they first arise (Section 6.1). The economic regulation controller replaces the quadratic state-tracking stage cost with a fuel-only cost while retaining the DARE terminal cost:

$$\min_U \sum_{i=0}^{N-1} u_i^\top R_{\text{eco}} u_i + x_N^\top P x_N \quad (15a)$$

$$\text{s.t. } x_{i+1} = A_d x_i + B_d u_i, \quad -u_{\max} \leq u_i \leq u_{\max}, \quad x_0 = x(k). \quad (15b)$$

This is the regulation MPC (13) with $Q_{\text{stage}} = 0$ and $R = R_{\text{eco}} = I_{n_u}$. The terminal cost P and gain K are computed from the DARE (9) with stabilising weights ($Q_{\text{dare}} \succ 0$, R_{dare}), $A_{\text{cl}} := A_d - B_d K$ Schur. Following [23], $V_f(x) = x^\top P x$ is a valid terminal cost for the economic stage cost provided the descent condition $P - A_{\text{cl}}^\top P A_{\text{cl}} \succeq K^\top R_{\text{eco}} K$ holds; using the DARE identity $P - A_{\text{cl}}^\top P A_{\text{cl}} = Q_{\text{dare}} + K^\top R_{\text{dare}} K$, this reduces to

$$M_1 := Q_{\text{dare}} + K^\top (R_{\text{dare}} - R_{\text{eco}}) K \succeq 0, \quad (16)$$

Condition (16) holds with margin at every sampling interval used in this report. Table 4 lists the eigenvalues of M_1 for the closing and docking scenarios. The QP matrices are

Table 4: Eigenvalues of M_1 (16) at each scenario's sampling interval

Scenario	Δt	$\text{eig}(M_1)$
Closing	120 s	$\{0.0100, 0.0100, 0.0100, 3.125, 3.127, 3.152\}$
Docking	10 s	$\{0.0104, 0.0104, 0.0104, 17.53, 17.53, 17.59\}$

$$H = 2(S_u^\top \bar{P} S_u + \bar{R}_{\text{eco}}), \quad f = 2 S_u^\top \bar{P} S_x x(k), \quad (17)$$

with $\bar{P} = \text{blkdiag}(\underbrace{0, \dots, 0}_{N-1}, P)$ and $\bar{R}_{\text{eco}} = I_N \otimes R_{\text{eco}}$.

At each sampling instant the QP is solved in error coordinates $e_k = x(k) - x_{\text{ref}}$ (the docking port x_{dock} in Operation 3, the handover point x_{switch} in Operation 1), the first input $u^*(k) = U_{1:n_u}^*$ is applied, the state is propagated one step, and the horizon recedes [3, 4]. During fly-around the chaser acquires and maintains a Natural Motion Circumnavigation (NMC). Its full state

$$x_{\text{nmc}}^*(t) = \begin{bmatrix} b \cos(\phi_0 + nt) \\ y_c - 2b \sin(\phi_0 + nt) \\ 0 \\ -b n \sin(\phi_0 + nt) \\ -2b n \cos(\phi_0 + nt) \\ 0 \end{bmatrix}, \quad (18)$$

has radial semi-axis b , initial phase ϕ_0 , along-track drift centre y_c , and mean motion n . Unlike the regulation controllers, this operation uses no terminal cost, because a soft terminal penalty alone leaves the target orbit unselected

under a pure-fuel cost. Instead the condensed QP keeps the fuel-only stage cost ($H = 2I_{n_u}N$, $f = 0$) and pins the orbit in one of two ways. Sampling the reference at the controller rate, $x_j^* = x_{\text{nmc}}^*(j \Delta t)$ for $j = 0, \dots, P-1$ with $P = T/\Delta t$ samples per period, the first variant imposes a phase-synchronised terminal equality on all six states,

$$x_N = x_{\sigma(k)}^*, \quad \sigma(k) = ((k + N - 1) \bmod P) + 1, \quad (19)$$

while the second replaces it with an ε -regularised phase-synchronised stage cost $\ell_{\text{aug}} = \|u\|^2 + \varepsilon\|x - x_k^*\|^2$ that recovers strict P -periodic dissipativity without any terminal constraint [24, 25, 26]. In both cases the phase index $\sigma(k)$ is a deterministic function of the global controller step k . The reference is phase-locked to mission time, advancing one sample per step and wrapping every P steps. The details are developed in Section 6.2.

5.4 Software Availability

The implementation developed for this work is available as an open-source repository under the BSD 3-Clause License:

<https://github.com/maiergamer9/empc-rpo-thesis>

The repository contains the controller implementation, simulation scripts, and supporting material required to reproduce the numerical studies presented in Section 6.

6 Mission Operations

6.1 Operation 1: Far-Range Closing

Operation 1 is the closing transfer from a 10 km V-bar hold to a handover box at 300 m, run at $\Delta t = 120$ s, $N = 20$ and with thrust bound $u_{\text{max}} = 10^{-2}$ m/s². The ADRIOS ConOps initiates rendezvous near 30 km and flies the far-range approach with ground-planned, impulsive burns under absolute navigation [13]. That segment is governed by orbital-mechanics targeting rather than closed-loop proximity control and lies outside the scope of this study. The present work models the close-proximity segment, which begins around 10 km once relative navigation is available and autonomous, continuously actuated MPC takes over. The actuator is accordingly modelled as a low-thrust device delivering a bounded acceleration $\|u\|_{\infty} \leq u_{\text{max}}$, held piecewise-constant over each sample by the zero-order hold, as opposed to the impulsive- Δv model of the far-range phase. The simulation stops as soon as the chaser enters a 50 m ball around the switch-point, i.e. when the full-state error satisfies $\|x_k - x_{\text{switch}}\|_2 < 50$ m. This suffices because Operation 2 does not require its initial condition to be exactly $x_0 = [0, 300, 0, 0, 0, 0]^T$. The horizon is capped at five orbital periods ($t_{\text{max}} = 5T$) in which every closing run converges well within, so five periods is the maximum admissible duration, not the realised one.

6.1.1 Dissipativity Gap and the Design Axis

The initial state $x_0 = [0, 10\,000, 0, 0, 0, 0]^T$ is itself a zero-input CW equilibrium. Under the pure fuel cost $\ell(x, u) = \|u\|^2$, the optimal first move is to do nothing and the closed loop never leaves 10 km. The value function remains bounded below by zero, but it is not positive definite with respect to the target. It attains its minimum, zero, at every V-bar equilibrium, including the 10 km initial point, rather than uniquely at the desired handover state. Equivalently, the system is not strictly dissipative with respect to the target regime [27]. No storage function renders the rotated stage cost positive definite, and the receding-horizon optimum is simply $u \equiv 0$. This is the core requirements finding for this operation. A pure fuel stage cost is insufficient for far-range closing rendezvous, and at least one additional structural ingredient (a terminal constraint, a terminal cost, or a state regularisation) is needed.

Closing this gap requires injecting the location and tail value of the desired steady state through a terminal ingredient. Three channels for doing so define the design axis for Operation 1:

- **Run A1:** A terminal ball constraint $\|C_p e_N\|_{\infty} \leq r$, with $r = 200$ m, that injects the target location (the switch-point) as a hard feasibility requirement, leaving the fuel stage cost untouched. Finite-time arrival of the predicted position into the handover box is observed in simulations.
- **Run B:** A stage-cost regularisation $\ell = \|u\|^2 + \rho_c \|e\|^2$, with $\rho_c = 10^{-11}$, that tilts the cost landscape so the target becomes the unique minimiser, with no offline computation. Yields practical asymptotic stability, backed by [23]’s stability result. The convergence neighbourhood shrinks as $N \rightarrow \infty$ or $\rho_c \rightarrow \infty$, but a larger ρ_c distorts the fuel objective.
- **Run C:** A CLF terminal cost $V_f(e) = e^T P e$ with $M_1 \succeq 0$ (§5.3), verified numerically at $\Delta t = 120$ s (Table 4). By [23], the descent property $M_1 \succeq 0$ certifies the asymptotic average-performance bound (the

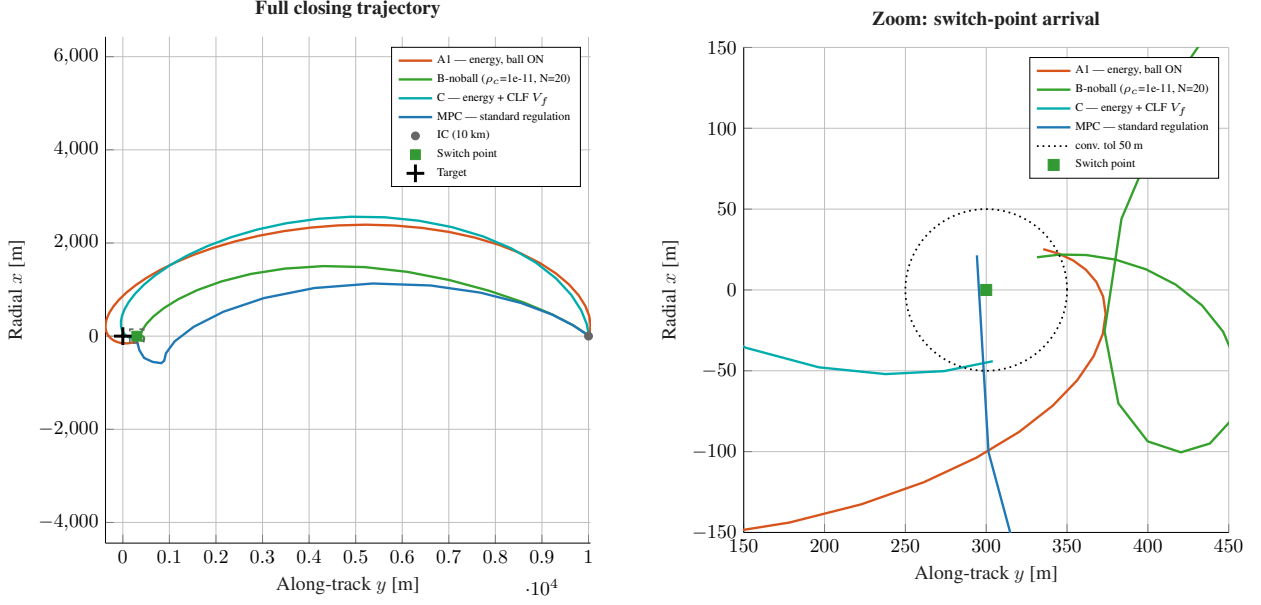


Figure 2: Closing trajectories — full view and switch-point zoom.

closed-loop running cost is asymptotically no worse than the optimal steady state) but it does not certify asymptotic stability of the target. Asymptotic stability would additionally require strict dissipativity, which again fails here for the same reasons already discussed in Section 6.1. Convergence to the target is observed in simulation and is therefore reported as an empirical result, not a proven guarantee.

6.1.2 Results

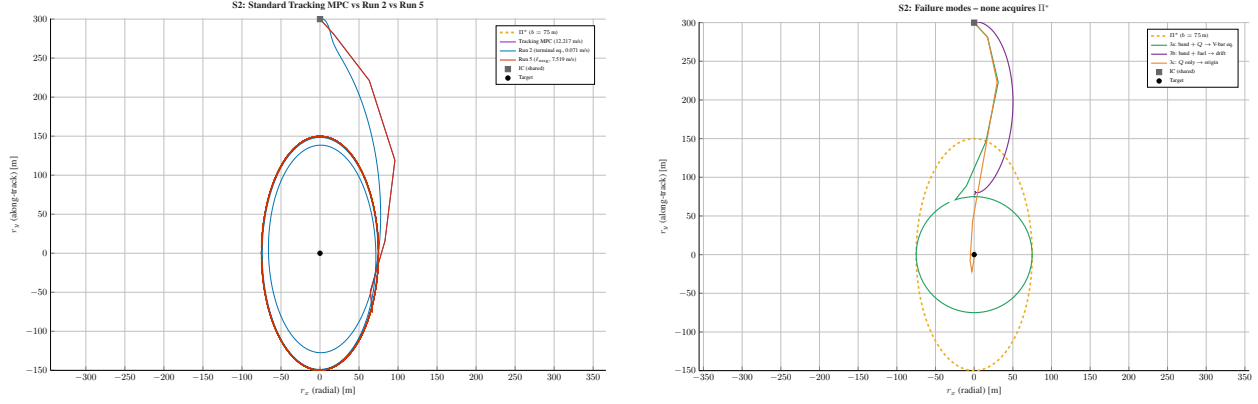
The three EMPC channels are compared against a Standard (quadratic-tracking) MPC with weights $Q_{\text{dare}} = \text{diag}(10^{-2}, 10^{-2}, 10^{-2}, 1, 1, 1)$ and $R_{\text{dare}} = \text{diag}(10^4, 10^4, 10^4)$. Table 5 and Fig. 2 report the outcome. All three channels resolve the $u \equiv 0$ degeneracy and complete the closing at a fraction of the Standard-MPC fuel cost. Standard MPC spends 31.72 m/s, whereas the economic variants range from 3.29 m/s (Run A1) to 18.33 m/s (Run B). The ranking $A1 < C < B$ mirrors how strongly each ingredient distorts the pure-fuel objective. The terminal ball (A1) leaves the stage cost untouched and is cheapest while the CLF terminal cost (C) adds only a terminal penalty and the stage-cost regularisation (B), which penalises state error at every step, is the most expensive economic option while still undercutting Standard MPC by 42 %. That the hard terminal ball is the least costly way to inject the target confirms the requirements reading of §6.1. The saving is bought with time. Driven by its state-tracking cost, Standard MPC converges fastest (21 steps, 0.67 h) by thrusting hard, whereas the fuel-optimal Run A1 coasts on the natural drift and takes three times as long (61 steps, 2.0 h). Runs B and C fall in between (≈ 1.2 h). This fuel-versus-time trade is the central engineering choice for the closing phase.

Table 5: S4 closing-approach run comparison.

Run	Δv [m/s]	Steps / time to conv.	term. distance [m]
Standard MPC	31.7196	21 / 0.67 h	22.17
Run A1 (EMPC + terminal ball)	3.2888	61 / 2.00 h	43.36
Run B (ρ_c regularisation)	18.3321	36 / 1.17 h	37.32
Run C (EMPC + CLF)	5.9800	37 / 1.20 h	44.29

6.2 Operation 2: NMC Fly-Around

Operation 2 acquires a 2:1 NMC fly-around orbit of semi-axis 75 m from the 300 m handover point, run at $\Delta t = T/92 \approx 60.27$ s, $N = 92$ (a full orbital period). The economic degeneracy here is sharper than in Operation 1. Because every passive NMC orbit is a zero-cost trajectory, pure fuel cost leaves the controller indifferent among the entire one-parameter family of ellipses, and it simply coasts on its natural drift ($\Delta v = 0$, no orbit acquired). Critically,



(a) Standard tracking MPC vs. the economic Run 2 (periodic terminal equality) and Run 5 (periodic ℓ_{aug}).

(b) Failure modes: neither a state band nor a state cost alone selects the NMC orbit.

Figure 3: Fly-around orbit acquisition and failure-mode comparison.

the periodic analogue of strict dissipativity also fails for the pure-fuel cost on this orbit family, so the rotated-cost route available in principle for economic MPC is closed here [24, 25]. A sequence of failure-mode runs (Table 6) establishes that neither a state cost nor a range band substitutes for the missing terminal ingredient. The instructive case is Run 3a. Q combined with a band manufactures a continuum of equal-cost non-equilibrium states and locks the QP into an expensive powered limit cycle, strictly worse than pure fuel alone. The ingredient that both selects the target orbit Π^* and fixes its phase is a hard periodic terminal equality $x_N = x_\varphi^*$ tying the horizon end to the phase-consistent point on the orbit.

Table 6: Operation 2 — outcome by cost / constraint combination.

Run	Cost	Constraint	Outcome
1	fuel	none	coasts, no orbit
2	fuel	periodic terminal eq.	captures Π^*
3c	Q	none	regulates to origin
3b	fuel	band	V-bar equilibrium
3a	Q	band	powered limit cycle
4	fuel	IC on Π^*	captures Π^*
5	phase-synchronised ℓ_{aug} ($\varepsilon = 10^{-4}$)	none	captures Π^*

The periodic terminal equality captures Π^* at total $\Delta v = 0.071$ m/s, with the target geometry. It discovered semi-axis $b \approx 74.9$ m against the 75 m target (range ratio 2.004), terminal control decaying to $\|u\| \rightarrow 3.6 \times 10^{-9}$ m/s² and last-orbit phase error at the sub-decimetres level. A regularised alternative (Run 5) drops the terminal constraint entirely, restoring strict periodic dissipativity through the phase-synchronised augmented cost ℓ_{aug} ($\varepsilon = 10^{-4}$), reaching the same orbit ($b = 75.0$ m, ratio 2.000) at total $\Delta v = 7.519$ m/s. The two routes converge to the same Π^* . The terminal equality is exact but binds the horizon to a full period and adds six equality rows. The regularised route frees the horizon and drops the constraint at the cost of a sub-cm/s fuel bias. Both implementations are compared to a standard Tracking MPC with $Q = I_6$, $R = I_3$ and terminal cost $P = \text{DARE}(A_d, B_d, Q, R)$, over a horizon $N_{\text{track}} = 20$.

Table 7: NMC orbit acquisition from the shared V-bar hold $x_0 = [0, 300, 0, 0, 0, 0]^\top$: standard tracking MPC versus the economic controllers. Capture = first time the phase error stays below 5 m.

Controller	Δv [m/s]	Saving vs. tracking	Capture [orbits]
Standard tracking MPC	12.217	—	0.08
Run 2 (terminal eq.)	0.071	99.4%	2.14
Run 5 (periodic ℓ_{aug})	7.519	38.5%	0.08

6.3 Operation 3: Final Docking

Operation 3 is the terminal docking approach from 150 m to the capture envelope, run at $\Delta t = 10$ s, $N = 24$, $u_{\max} = 10^{-2}$ m/s², from a V-bar forced-keep-point initial condition, with convergence declared at $\|e_r\| < 1$ m on the position error. An appropriate distance for switching from GNC to the capture mechanism. A pure-fuel stage cost would again yield the degeneracy described in 6.1 and 6.2. Thus, Operation 3 again implements an EMPC with terminal cost term and a Standard MPC, with the same weight matrices as in Operation 1. Instead of a regularised stage-cost or a terminal ball, the objective of this phase admits a hard terminal equality constraint $x_N = x_{\text{dock}}$. All three are compared in Δv , convergence time and terminal distance.

Neither the 150 m start nor the horizon $N = 24$ is a free parameter and both are imposed by the upstream mission. The fly-around of Operation 2 departs at the V-bar crossing of the natural motion circumnavigation, which for a radial semi-axis $b = 75$ m lies at $y = 2b = 150$ m. This geometry in turn sets a hard requirement on the docking controller. The terminal-equality constraint $x_N = x_{\text{dock}}$ demands that the origin be reachable in both position and velocity within the horizon. The minimum feasible horizon was established empirically. N was swept downward in simulation until the docking QP first became infeasible in the first steps, at which point the controller silently degrades to a best-effort, constraint-violating fallback. $N = 24$ is the minimum feasible horizon found this way, and is used as the common horizon for all three runs. A closed-form derivation of this bound for the coupled, constrained CWH dynamics is left as future work. The requirement binds only Run 3. Runs 1 and 2 replace the equality with a terminal cost and stay feasible at any N , but since a common horizon is used, $N = 24$ becomes the design point, fixed by the most demanding controller. This is a concrete instance of a mission-derived MPC requirement.

The approach is deliberately confined to an along-track corridor. In particular a line-of-sight cone of half-angle 30° about the +V-bar axis, together with a $y \geq 0$ half-space, constrains the chaser to close on the port from the along-track direction. This is the geometry preferred for docking-port alignment and capture-sensor line-of-sight, and it keeps the chaser out of the radial band from which a dispersion would carry it across the port. Constrained MPC for the terminal docking approach has been studied extensively [5, 6, 7]. Table 8 reports the three-run comparison and Fig. 4 shows the Hill-frame trajectories with the corridor shaded.

Table 8: S1 docking-approach run comparison (150 m, $N = 24$).

Run	Δv [m/s]	Steps / time to conv.	Term. distance [m]
Run 1 – Standard MPC	2.451	24 / 4.00 min	0.948
Run 2 – EMPC + CLF	1.314	43 / 7.17 min	0.988
Run 3 – EMPC + hard terminal equality	1.322	44 / 7.33 min	0.691

EMPC roughly halves the fuel cost relative to Standard MPC (1.31 m/s vs. 2.45 m/s, a 46.4% saving) at the cost of a longer settling time (43 vs. 24 steps, 7.2 min vs. 4.0 min). Runs 2 and 3 are near-identical in Δv (1.31 m/s vs. 1.32 m/s). Since the V-bar hold is a CW equilibrium, the CLF terminal cost P_{clf} already acts as an implicit soft terminal equality, so the explicit constraint of Run 3 adds little fuel. Its effect is instead on the terminal distance, which it tightens from 0.99 m to 0.69 m.

Declaring convergence at the 1 m capture scale is also what keeps the corridor constraints well-posed. The trajectories halt below 1 m, upstream of the cone apex and the $y = 0$ face, where the cone and half-space would otherwise become singular.

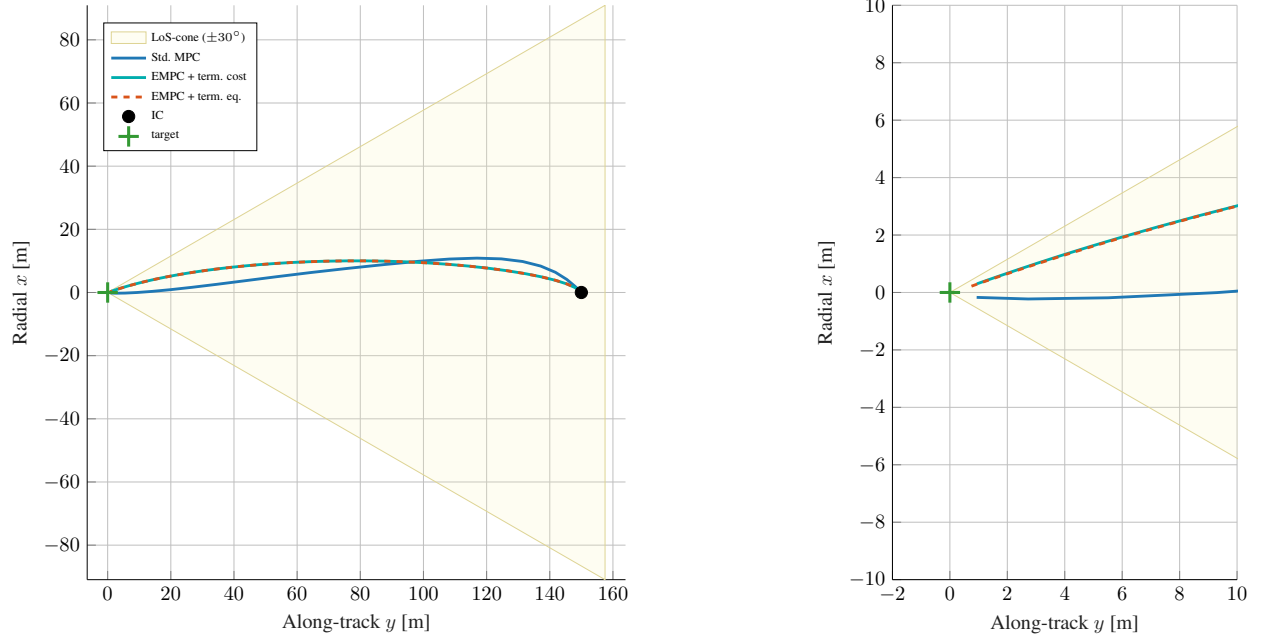


Figure 4: Hill-frame docking trajectory (radial x vs. along-track y). Left: full approach from the 150 m V-bar hold to the port; right: terminal zoom (≤ 10 m).

7 Cross Table

The three operations were analysed separately, each with its own dissipativity gap, terminal ingredient, and certificate. Table 9 first combines them end-to-end, giving the total fuel and time cost of running the mission with every controller choice per operation, so the cumulative effect of stacking ingredients across the sequence is visible in one place. Table 10 then draws the requirements analysis itself together. For each operation it lines up the mission requirement that motivated the ingredient, what that ingredient formally guarantees, what was actually observed in simulation, and the residual limitation that guarantee does not cover.

Table 9: End-to-end mission cost for every controller combination (Closing $\times$ Fly-around $\times$ Docking). Total Δv excludes the fixed departure burn; duration is for the complete three-operation sequence, computed from the respective initial conditions of each operation.

#	Closing	Flyaround	Docking	Δv_{tot} [m/s]	Duration [h]
1	energy+ball (A1)	periodic-eq (R2)	EMPC-CLF (R2)	4.7692	11.01
2	energy+ball (A1)	periodic-eq (R2)	EMPC-termeq (R3)	4.7771	11.01
3	energy+ball (A1)	augmented (R5)	EMPC-CLF (R2)	11.0724	11.01
4	energy+ball (A1)	augmented (R5)	EMPC-termeq (R3)	11.0809	11.01
5	dissipative (B)	periodic-eq (R2)	EMPC-CLF (R2)	19.9102	10.14
6	dissipative (B)	periodic-eq (R2)	EMPC-termeq (R3)	19.9103	10.14
7	dissipative (B)	augmented (R5)	EMPC-CLF (R2)	26.5352	10.14
8	dissipative (B)	augmented (R5)	EMPC-termeq (R3)	26.5431	10.15
9	energy+CLF (C)	periodic-eq (R2)	EMPC-CLF (R2)	7.6459	10.21
10	energy+CLF (C)	periodic-eq (R2)	EMPC-termeq (R3)	7.6539	10.21
11	energy+CLF (C)	augmented (R5)	EMPC-CLF (R2)	13.4763	10.21
12	energy+CLF (C)	augmented (R5)	EMPC-termeq (R3)	13.4842	10.21

Table 10: Cross-scenario requirements synthesis: for each operation, the mission requirement that drives the MPC ingredient, the theoretical property claimed for it in the cited EMPC literature, the observed result, and the residual limitation. Baseline (quadratic) controllers in grey for contrast. The entries in the “Claimed property” column are not re-derived in this report. They state the guarantee associated with the cited result under its stated hypotheses, a system-specific verification of which is left as future work (Section 8).

Op	Ingredient (Run)	Mission req.	Claimed property	Observed result	Limitation
1	Quadratic tracking (Std MPC)	Reach box around 300 m switch point	Asymptotic stability (LQR)	$\Delta v = 31.7$ m/s, 0.67 h	Fights drift; 5–10× EMPC fuel
	Pure fuel + ℓ_∞ terminal ball, $r = 200$ m (A1)		Predicted arrival into box (feasibility); no stability cert.	$\Delta v = 3.29$ m/s, 2.0 h (best fuel)	Slowest; feasibility $\neq$ stability
	Pure fuel + $\rho_c \ e\ ^2$ regularisation (B)		Practical asymptotic stability	$\Delta v = 18.3$ m/s, 1.17 h	ρ_c distorts fuel; underlying stabilisability hypothesis not verified here
	Pure fuel + CLF terminal cost, $M_1 \geq 0$ (C)		Avg-performance bound (<i>not</i> asymptotic stability)	$\Delta v = 5.98$ m/s, 1.2 h	Strict dissipativity fails; convergence empirical
2	Quadratic tracking (Std MPC)	Acquire NMC orbit	Asymptotic stability	$\Delta v = 12.2$ m/s, cap. 0.08 orb.	Fights free motion; high fuel
	Pure fuel + periodic terminal eq. $x_N = x_{\sigma(k)}^*$ (Run 2)		Recursive feasibility + avg-performance bound; no orbit-stability cert.	$\Delta v = 0.071$ m/s (99.4%), $b \approx 74.9$ m	Horizon bound to full period $N = P = 92$; slow capture (2.1 orb.); hypotheses not re-verified for this system
	Pure fuel + $\varepsilon \ x - x_k^*\ ^2$ phase-sync (Run 5)		P -periodic strict diss. claimed by cited result (constancy of ℓ , V_N on Π^* assumed, not checked here) $\Rightarrow$ practical stability	$\Delta v = 7.52$ m/s (38.5%), $b = 75.0$ m	Sub-cm/s fuel bias; not exact
3	Quadratic tracking + DARE (Std MPC)	Reach docking-port vicinity (<1 m)	Asymptotic stability	$\Delta v = 2.45$ m/s, 24 steps	Highest docking fuel; saturated
	Pure fuel + CLF terminal cost (Run 2)		Avg-performance bound	$\Delta v = 1.31$ m/s (46%), term. 0.99 m	No asymptotic-stability certificate
	Pure fuel + hard terminal eq. $x_N = x_{\text{dock}}$ (Run 3)		Exact terminal state (feasibility); $N = 24$ found empirically	$\Delta v = 1.32$ m/s, term. 0.69 m (tightest)	$N = 24$ from sim. sweep; analytical derivation is future work

8 Conclusion

This report developed and applied a mission-centric EMPC design framework to three sequential proximity operations of the ADRIOS / ClearSpace-1 mission. For each operation, the same condensed-QP formulation was instantiated with a pure-fuel stage cost, the resulting dissipativity gap was identified analytically, and a terminal ingredient (a terminal region, a terminal cost, a terminal equality, or a stage-cost regularisation) was selected and motivated against the operation’s specific mission requirement. The three instances were then drawn together into a single cross-scenario table relating mission requirement, MPC ingredient, expected theoretical property, and residual limitation. This is the first concrete step toward the traceable requirements-to-design mapping motivated in Section 1.

The analysis rests on a number of simplifying assumptions that should be kept in mind when interpreting the results.

The relative dynamics are linearised CWH, valid only for near-circular reference orbits and small relative separations, and neglect J_2 , drag, and solar radiation pressure. The chaser is modelled as a point mass with an idealised, continuously throttleable low-thrust actuator; no attitude dynamics, 6-DOF pose tracking, or actuator discretisation are represented, and the proximity-manoeuving phase (FKP $\rightarrow$ FAP) is excluded entirely (Section 2). Full, noise-free state feedback is assumed at every sampling instant, so no navigation or model uncertainty enters the design, and the quadratic control-effort cost $\|u\|^2$ is used throughout as a convex surrogate for the true Δv objective $\|u\|_1$. Most importantly, the stability and performance properties attributed to each terminal ingredient in Table 10 are drawn directly from the corresponding results in the EMPC literature rather than re-derived here for the specific CWH system and constraint sets used in this report; no certificate-by-certificate proof is provided, and the reported “guarantees” should be read as the applicable condition under which the cited theorem holds, not as a self-contained derivation.

Several of these simplifications point directly to open work. Foremost among them, a rigorous, system-specific derivation of the stability and performance guarantees sketched in Table 10, verifying the exact hypotheses of the cited theorems against the CWH dynamics, constraint sets, and terminal ingredients used in each operation, is left as future work. Extending the framework to the proximity-manoeuving and capture phases would require 6-DOF pose-tracking MPC. Treating navigation and model uncertainty would move the guarantees from nominal to robust or stochastic MPC. And a mission-wide storage function would be needed to reason formally about stability across the full switched sequence rather than within each phase in isolation. The minimum feasible horizon for the Operation 3 terminal equality ($N = 24$, Section 6.3) was established by a simulation sweep rather than a closed-form bound for the coupled, constrained dynamics; deriving such a bound analytically is left as future work. The passive-safety constraint introduced in Section 2.2 is not yet evaluated as a dedicated mission requirement, and remains an open item for the next iteration of this report. Finally, all controllers were validated only in simulation with quadprog.

Author Contributions

Nils Maier: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – Original Draft. **Torbjørn Cunis:** Conceptualization, Resources, Writing – Review & Editing, Supervision, Project administration. **Tam W. Nguyen:** Methodology, Resources, Writing – Review & Editing, Supervision, Project administration.

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